USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.EPR.SNK3.14 :Wait and SoftReset in EPR
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
New test to verify EPR_Mode enables properly KeepAlive even is still in SPR contract due to Wait on first EPR_Request. New test checks also DUT behavior on SoftReset while in EPR contract.

Benefits as a result of the proposed changes:
Check main specification requirements

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

New:

Test.PD.EPR.SNK3.14: Wait and SoftReset Test in EPR

Description: The Tester as a Source checks that the UUT properly initialize KeepAlive process while in EPR Mode but in SPR contract due to Wait on EPR_Request. Tester checks also that SoftReset during EPR mode is properly handledby DUT.

	Consumer Only	Provider Only	DRP, C/P, P/C
Rev3ChkdSnk	✓		✓

Check Applicability: The Test is applicable if the VIF Parameter EPR_Supported_As_Snk is set to Yes
Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
    b. The UUT has VIF field PD_Port_Type set to Consumer Only, Consumer /Provider or DRP, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2

2. The tester waits for Time_To_EPR_Ready maximum to confirm following checks: [TEST.PD.EPR.SNK3.14#1]
    a. If the "EPR mode capable" field is set to 0 in the *Request* Message sent by UUT during the bring up procedure, the test fails if no other Request is received with “EPR mode capable” field is set and the test stop here
    b. The test fails if the UUT does not send an *EPR_Mode* Enter message with the Action field set to 0x01 and the test stops here. [TEST.PD.EPR.SNK3.14#2]

3. The tester sends the *EPR_Mode* Message with the Action field set to 0x02(Enter Acknowledged) and the Data field set to 0x0
4. If VIF field Captive_Cable = Yes, the Tester sends SOP' *Discover Identity* Command and checks the below fields in the response message [TEST.PD.EPR.SNK3.14#3]
    a. V_{BUS} Current Handling Capability set to 10b(5A)
    b. Maximum V_{BUS} Voltage set to 11b(50V)
    c. EPR Mode Capable field set to 1b (EPR Mode Capable)

5. The Tester sends the *EPR_Mode* Message with the Action field set to 0x03(Enter_Succeeded) and the Data field set to 0x0 within *tEnterEPR* from the *EPR_Mode* (Enter) Message.

6. The Tester sends the *EPR_Source_Capabilities* Message that contains seven PDOs and four EPR_PDOs (240W)
    a. B31…30 (Fixed Supply) set to 00b.
    b. B29 (Dual-Role Power) set to 0b if Tester is a Source only; set to 1b if Tester is a Dual-Role Power
    c. B28 (USB Suspend Supported) set to 0b.
    d. B27 (Unconstrained Power) set to 1b.
    e. B26 (USB Communications Capable) set to 0b.

f. B25 (Dual-Role Data) set to 0b.
g. B24 (Unchunked Extended messages Supported) set to 0b or 1b depending on test condition.
h. 23 set to 1b (*EPR_Mode* Capable)
i. 22 (Reserved) set to 0b
j. B21…20 (Peak Current) set to 00b.

PDO1: 9. B19…10 (Voltage) set to 5V.
10. B9…0 (Maximum Current) set to 3000mA
PDO2: 9. B19…10 (Voltage) set to 9V.
10. B9…0 (Maximum Current) set to 3000mA
PDO3: 9. B19…10 (Voltage) set to 15V.
10. B9…0 (Maximum Current) set to 3000mA
PDO4: 9. B19…10 (Voltage) set to 20V.
10. B9…0 (Maximum Current) set to 5000mA
PDO5: B31..30 (APDO) set to 11b
B29..28 (SPR AVS) set to 10b
B27..26 (Peak Current) set to 00b
B25..20 (Reserved) set to 00b
B19..10 (9-15V current) set to 3000mA
B9..0 (15-20v current) set to 5000mA
PDO6: All fields set to 0.
PDO7: All fields set to 0.
PDO8: 9. B19…10 (Voltage) set to 28V.
10. B9…0 (Maximum Current) set to 5000mA
PDO9: 9. B19…10 (Voltage) set to 36V.
10. B9…0 (Maximum Current) set to 5000mA
PDO10: 9. B19…10 (Voltage) set to 48V.
10. B9…0 (Maximum Current) set to 5000mA
PDO11: 1. B31…B30 (Augmented) set to 11b.
2. B29..B28 (EPR Adjustable Voltage Supply) 01b
3. B27..B26 (Peak Current) set to 00b
4. B25..B17(maximum voltage) set to 48V
5. B16(reserved) set to 0b
6. B15..B8 (minimum voltage) set to 15V
7. B7..B0 (PDP) set to 240W

7. The Tester check fails, and the test stops here if the UUT does not respond with the *EPR_Request* Message [TEST.PD.EPR.SNK3.14#4]

8. The Tester sends the *Wait* Message.

9. The Tester checks that the UUT does not send Hard Reset signaling for at least *tProtErrHardReset* (this timing is measured from the last bit of the *GoodCRC* of the *EPR_Request* Message) [TEST.PD.EPR.SNK3.14#5]

10. If there are no messages initiated by the UUT for more than *tSinkEPRKeepAlive* max (this timing is measured from the last bit of the *GoodCRC* of the *Wait* Message), then the check fails [TEST.PD.EPR.SNK3.14#6]

11. The Tester sends *EPR_Get_Sink_Cap* Message and intentionally never sends the *GoodCRC* upon the first chunk of *EPR_Sink_Capabilities* Message reception. The test fails if DUT does not retry for *nRetryCount* times the first chunk of *EPR_Sink_Capabilities* Message[TEST.PD.EPR.SNK3.14#7]

12. The Tester checks if the UUT transmits a *Soft Reset* Message within *tSoftReset* of *tReceive* expiring. The check fails if the *Soft Reset* Message is not received within the following time interval [*tReceive* min, *tReceive* max + *tSoftReset* max] or [0.9ms, 16.1ms]. [TEST.PD.EPR.SNK3.14#8]. The Tester responds with a valid *GoodCRC*.

13. The tester sends *Accept* Message and establishes EPR contract using common procedure COMMON.PROC.PD3.3 from Step2. [TEST.PD.EPR.SNK3.14#9]

14. The Tester waits for 600 milliseconds and if there are no messages initiated by the UUT for more than *tSinkEPRKeepAlive* max (this timing is measured from the last bit of the *GoodCRC* of the *PSRDY* Message), then the check fails [TEST.PD.EPR.SNK3.14#10]